

Semi-Supervised Echocardiography Video Segmentation via Adaptive Spatio-Temporal Tensor Semantic Awareness and Memory Flow

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Abstract—Accurate segmentation of cardiac structures in echocardiography videos is vital for diagnosing heart disease. However, challenges such as speckle noise, low spatial resolution, and incomplete video annotations hinder the accuracy and efficiency of segmentation tasks. Existing video-based segmentation methods mainly utilize optical flow estimation and cross-frame attention to establish pixel-level correlations between frames, which are usually sensitive to noise and have high computational costs. In this paper, we present an innovative echocardiography video segmentation framework that exploits the inherent spatio-temporal correlation of echocardiography video feature tensors. Specifically, we perform adaptive tensor singular value decomposition (t-SVD) on the video semantic feature tensor within a learnable 3D transform domain. By utilizing learnable thresholds, we preserve the principal singular values to reduce redundancy in the high-dimensional spatio-temporal feature tensor and enforce its potential low-rank property. Through this process, we can capture the temporal evolution of the target tissue by effectively utilizing information from limited labeled frames, thus overcoming the constraints of sparse annotations. Furthermore, we introduce a memory flow method that propagates relevant information between adjacent frames based on the multi-scale affinities to precisely resolve frame-to-frame variations of dynamic tissues, thereby improving the accuracy and continuity of segmentation results. Extensive experiments conducted on both public and private datasets

validate the superiority of our proposed method over state-of-the-art methods, demonstrating improved performance in echocardiography video segmentation.

Index Terms—Echocardiography video, low-rank property, semi-supervised segmentation, spatio-temporal correlation.

I. INTRODUCTION

ECHOCARDIOGRAPHY has been widely used to assess cardiac function and diagnose cardiovascular disease due to its real-time capabilities, cost-effectiveness, and non-invasive nature [1]. By segmenting the cardiac tissues, doctors can obtain detailed information about the anatomy of the heart, aiding in detecting abnormalities and localizing diseased areas. Currently, manual segmentation is considered the gold standard for analyzing echocardiography videos. However, it is time-consuming, subjective, and prone to inter- and intra-observer variability. Consequently, there is an urgent need to develop automated segmentation methods for echocardiography videos to improve efficiency and reliability in clinical practice.

Echocardiography video segmentation faces several challenges that hinder its accuracy and efficiency [2]. One of the primary challenges arises from the physical properties of ultrasound imaging, which introduce speckle noise into the acquired images [3]. Speckle noise appears as spots of varying brightness within a frame, leading to significant degradation in image quality and making image segmentation difficult. In addition, compared to other imaging modalities such as magnetic resonance imaging (MRI), X-ray, and computed tomography (CT), ultrasound images typically exhibit lower spatial resolution and contrast [4]. Consequently, this leads to challenges in accurately identifying and delineating the boundaries of segmentation targets. Furthermore, echocardiography videos usually consist of multiple frames, varying in quantity from tens to hundreds. However, due to the expensive cost of manual annotation, physicians typically only annotate the end-diastolic (ED) and end-systolic (ES) frames within a single cardiac cycle, resulting in the limitation of sparse annotations. It is imperative to tackle these challenges to enhance the precision of echocardiography video segmentation.

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Traditional methods [5], [6], [7] typically utilize the region growing method [8], active contour model [9], and K-means clustering algorithm [10] to segment ultrasound images. However, these methods are known to be sensitive to noise and often result in poor segmentation results. In recent years, deep learning techniques have shown promising results in medical image segmentation tasks. Most existing methods for echocardiography video segmentation mainly utilize 2D convolutional neural networks to perform frame-by-frame segmentation [11], [12], [13]. Zhou et al. [14] proposed a dual-branch approach that leverages a shared encoder for feature extraction and two decoders for the segmentation of myocardium and left ventricle, respectively. Liu et al. [15] proposed using a pyramid local attention module to capture multi-scale information, enhancing the ability of feature extraction and improving the segmentation performance. However, these single-frame methods ignore the temporal correlation between frames, affecting the accuracy and continuity of video segmentation results. Furthermore, these methods require a large number of labeled frames for training, making it difficult to obtain satisfactory results in sparsely annotated echocardiography video segmentation tasks.

To address the limitations, recent methods have attempted to explore the relationships between frames to fuse features of adjacent frames [16], [17], [18], [19], [20]. For instance, Bai et al. [21] proposed a combination of fully convolutional networks and recurrent neural networks to incorporate spatial and temporal information into the segmentation task. They also introduced non-rigid label propagation to solve the issue of sparse video annotations. Similarly, Wu et al. [22] proposed a semi-supervised echocardiography video segmentation method that aligned and fused features of consecutive frames via the spatio-temporal semantic calibration module. However, these methods usually only consider partial adjacent frames and neglect to take the video data as a whole to fully utilize the temporal profile information.

Some methods collaborate with optical flow learning to fully exploit temporal information to track dynamic targets in videos [23], [24], [25]. For instance, Wei et al. [26] proposed a 3D U-Net method consisting of two decoders that can simultaneously estimate the cardiac shape and motion to collaboratively segment and track target tissues, enforcing the temporal consistency of segmentation results. Li et al. [27] utilized the Conv-LSTM units based on an optical flow motion estimation module to analyze video spatio-temporal consistency and improve segmentation accuracy. However, due to the lack of corresponding labels, the optical flow modules in such methods are trained in an unsupervised learning manner, making it difficult to obtain accurate results. In addition, problems such as low signal-to-noise ratio, speckle noise, and limited contrast in ultrasound images often lead to errors in pixel-level optical flow prediction. The absence of mid-frame segmentation labels further exacerbates these problems. To improve the accuracy of segmentation results, it is crucial to explore novel prior principles of echocardiography videos as guiding information for semi-supervised segmentation.

Recent advances in SAM-based methods have sparked interest in adapting SAM [28] for medical image

segmentation [29], [30], [31], [32], [33], [34]. For instance, MedSAM [35] preserves SAM's architecture while incorporating bounding box prompts specific to the medical domain and focuses on fine-tuning the mask decoder. In contrast, SAM-Med2D [36] and SAMUS [37] modify the image encoder to better handle medical images. SAM-Med2D introduces a learnable adapter layer into the encoder. SAMUS introduces position and feature adapters to improve performance on a large ultrasound image dataset. Despite these advancements, most SAM-based methods are designed for single-frame image segmentation and can not effectively handle video sequences. MemSAM [38] extends SAM by integrating a memory enhancement module, improving performance in semi-supervised video segmentation. SAM2 [39] incorporates a streaming memory that stores previous prompts and predictions to interactively segment regions using prompts on video frames. Medical SAM2 [40] extends SAM2 to the medical imaging domain by introducing a self-sorting memory bank and a weighted resampling strategy. However, the high computational cost of SAM remains a challenge for video data, significantly increasing inference and training time.

During one cardiac cycle, echocardiography can acquire dozens of frames that depict the progressive evolution of the heart's anatomy from relaxation to contraction. Adjacent frames in echocardiography videos often contain shared tissues and features that exhibit minimal differences. When using the same encoder for each frame to extract deep semantic features, the features corresponding to pixels of the same category are supposed to be situated in the same feature space. This implies the existence of potential spatio-temporal semantic correlation between consecutive frames. Based on this observation, we propose to utilize low-rank tensors to characterize the high-dimensional features of echocardiography videos, which assists in reducing feature redundancy while highlighting key semantic information.

Low-rank approximations are widely used in video restoration [41], dynamic MR image reconstruction [42], hyperspectral image denoising [43], and other fields [44], [45], [46], [47] to process videos or image sequences. However, their usage in semantic segmentation tasks remains relatively limited. In recent research, Pan et al. [48] proposed using the collective matrix factorization [49] method to decompose feature maps from multiple views of the same image into a shared dictionary matrix and multiple low-rank coefficient matrices. This method aims to obtain robust semantic representation across views in the self-supervised single image segmentation task, thereby improving the segmentation accuracy. However, the dictionary matrix needs to be initialized and learned iteratively, which increases the complexity of the method. Chen et al. [50] introduced a learnable tensor canonical-polyadic (CP) [51] decomposition module for supervised single-image segmentation. This module generates 3D attention maps of an image in the height, width, and channel directions, reducing the computational complexity involved in high-dimensional similarity calculations. Nevertheless, the rank of CP decomposition remains a hyperparameter that requires careful manual tuning. Despite the achievements of these previous methods in low-

rank approximations, to the best of our knowledge, how to use low-rank tensor approximation to represent video features for dense prediction tasks has hardly been explored.

In this paper, we present an innovative semi-supervised echocardiography video segmentation method, termed SAMF-Net, which incorporates two key modules: the adaptive spatio-temporal tensor semantic awareness (ASTSA) module and the memory flow (MF) module. Compared to current video segmentation methods that mainly rely on pixel-level correlations between frames, our approach explores semantic correlations across the video using spatio-temporal tensor decomposition techniques. Concretely, in the ASTSA module, we perform adaptive tensor singular value decomposition (t-SVD) on the video feature tensor and preserve the principal singular values via learnable truncation thresholds to remove redundant features, thereby enforcing the low-rank property of the video feature tensor. The ASTSA module can model the overall temporal evolution of dynamic tissues throughout the video to overcome the limitation of sparse annotation in echocardiography video segmentation. Moreover, we propose the MF module to handle the detailed changes of tissues between adjacent frames. In the MF model, we calculate cross-frame affinities at multiple scales to effectively propagate both semantic and mask information from the previously segmented frame to the subsequent frame. This enhances the feature representation of dynamic tissues and improves the accuracy of echocardiography video frame segmentation. To assess the efficacy of our proposed method, we conducted extensive experiments on both public and private datasets. The results demonstrate that our approach outperforms the current state-of-the-art methods by providing more accurate segmentation results.

The main contributions of this work can be summarized as follows,

- 1) We present a novel semi-supervised segmentation framework for echocardiography videos to improve the performance of segmentation in scenarios with sparse annotation. Unlike existing video segmentation approaches that rely on pixel-level correlation, we exploit video-level spatial-temporal correlation within the feature space.
- 2) We propose to utilize adaptive t-SVD on the high-dimensional video semantic feature tensor in a learnable 3D transform domain to enforce the inherent low-rank property of video features, leading to a compact and continuous representation of video features even with limited annotation.
- 3) To enhance the feature representation of dynamic tissues, we propose the memory flow module that models non-local pairwise affinity to propagate relevant information between adjacent frames, further improving the segmentation accuracy.
- 4) We conduct extensive experiments for left ventricle segmentation of echocardiography videos on two public datasets and a private dataset. The results demonstrate the superiority of our proposed method over existing state-of-the-art methods.

II. PRELIMINARIES: TENSOR LOW-RANK FACTORIZATION

In this section, we provide an overview of the fundamental concepts related to tensor low-rank factorization [52], [53].

For a three-dimensional tensor $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, we use $\mathcal{X}^{(n)}$ to represent the n -th frontal slice of $\mathcal{X}$, which can be denoted as $\mathcal{X}(:, :, n)$ for $n = 1, 2, \dots, d_3$. Given a transformation $\mathfrak{S} : \mathbb{R}^{d_1 \times d_2 \times d_3} \rightarrow \mathbb{R}^{e_1 \times e_2 \times e_3}$ and its inverse transformation $\tilde{\mathfrak{S}} : \mathbb{R}^{e_1 \times e_2 \times e_3} \rightarrow \mathbb{R}^{d_1 \times d_2 \times d_3}$, we can obtain $\mathcal{X}_{\mathfrak{S}} = \mathfrak{S}(\mathcal{X})$ and $\mathcal{X} = \tilde{\mathfrak{S}}(\mathcal{X}_{\mathfrak{S}})$.

The block diagonal matrix based on the frontal slices of $\mathcal{X}_{\mathfrak{S}}$ is defined as follows,

$$\overline{\mathcal{X}_{\mathfrak{S}}} = bdiag(\mathcal{X}_{\mathfrak{S}}) = \begin{bmatrix} \mathcal{X}_{\mathfrak{S}}^{(1)} & & & \\ & \mathcal{X}_{\mathfrak{S}}^{(2)} & & \\ & & \ddots & \\ & & & \mathcal{X}_{\mathfrak{S}}^{(e_3)} \end{bmatrix}. \quad (1)$$

We can utilize the fold operator to convert $\overline{\mathcal{X}_{\mathfrak{S}}}$ back into a tensor, which is described as follows,

$$fold(\overline{\mathcal{X}_{\mathfrak{S}}}) = fold(bdiag(\mathcal{X}_{\mathfrak{S}})) = \mathcal{X}_{\mathfrak{S}}. \quad (2)$$

Definition 1 ($\mathfrak{S}$ - Product): Given tensor $\mathcal{A} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, tensor $\mathcal{B} \in \mathbb{R}^{d_2 \times d_4 \times d_3}$, and transformation $\mathfrak{S}$, the $\mathfrak{S}$ -product is defined as follows,

$$\mathcal{C} = \mathcal{A} *_{\mathfrak{S}} \mathcal{B} = \tilde{\mathfrak{S}}(fold(\overline{\mathcal{A}_{\mathfrak{S}}} \times \overline{\mathcal{B}_{\mathfrak{S}}})) , \quad (3)$$

where the operator “ $\times$ ” means the standard matrix product.

According to (1), (2) and (3), we can obtain the following,

$$\overline{\mathcal{C}_{\mathfrak{S}}} = \overline{\mathcal{A}_{\mathfrak{S}}} \times \overline{\mathcal{B}_{\mathfrak{S}}}, \quad (4)$$

which means that the $\mathfrak{S}$ - product can be obtained by performing slice-wise matrix multiplication on each frontal slice of $\mathcal{A}_{\mathfrak{S}}$ and $\mathcal{B}_{\mathfrak{S}}$ in the transform domain.

III. METHODOLOGY

A. Overview

In this study, we propose a novel end-to-end framework called SAMF-Net for semi-supervised segmentation of echocardiography videos, as shown in Fig. 1. The SAMF-Net is built upon the encoder-decoder architecture and consists of three main modules: the multi-scale fusion (MSF) module, the adaptive spatio-temporal tensor semantic awareness (ASTSA) module, and the memory flow (MF) module.

Given an echocardiography video of size $H \times W \times T$, where H and W represent the spatial resolutions of each frame, and T is the number of frames, we utilize a shared encoder to extract deep semantic features $F_t \in \mathbb{R}^{C \times h \times w}$, $t = 1, 2, \dots, T$ for each frame. Here, h and w denote the spatial resolutions of the features, and C is the number of channels. Subsequently, we stack the features F_t in time order to form a high-dimensional video feature tensor $\mathcal{X}_{in} \in \mathbb{R}^{C \times h \times w \times T}$, which is then processed by the ASTSA module. In the ASTSA module, to exploit the spatial-temporal semantic correlation of the feature tensor, we perform adaptive t-SVD on $\mathcal{X}_{in}$, obtaining a compact and continuous video feature representation denoted as $\mathcal{X}_{out}$. We then separate $\mathcal{X}_{out}$ into features of each frame in time order, represented as F'_t . For the first frame, we input the feature F'_1 into a shared decoder to generate the segmentation result M_1 . Starting from the second frame, in order to effectively handle delicate changes

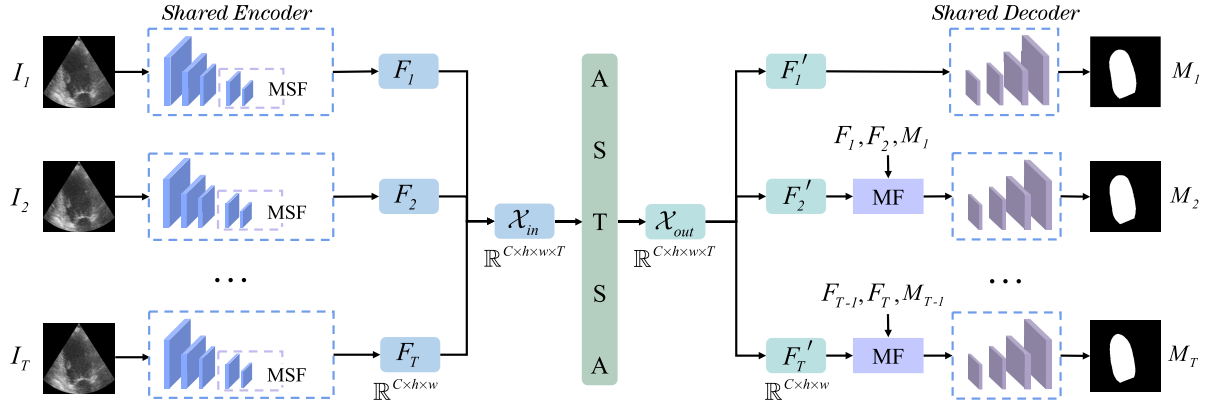


Fig. 1. Our proposed semi-supervised echocardiography video segmentation method (SAMF-Net) mainly consists of a shared encoder, a shared decoder, a multi-scale fusion (MSF) module, an adaptive spatio-temporal tensor semantic awareness (ASTSA) module, and a memory flow (MF) module.

between consecutive frames of dynamic tissues, we employ the MF module. This module calculates pixel-level affinities to propagate relevant information from the previously segmented frame to the next frame to be segmented. The MF module takes features F_{t-1} , F_t , and F'_t , as well as the segmentation result M_{t-1} of the previous frame as inputs, and then passes the output to the decoder to obtain the segmentation result M_t of the t -th frame.

In the encoder, we extract features at multiple scales and integrate information from the last two scales using the MSF module. Firstly, we resize the high-scale features to match the size of the low-scale features. Then, we divide and concatenate feature maps from both scales into four groups along the channel dimension. To fuse these features, different groups employ convolutional layers with varying receptive fields, including 1×1 , 3×3 , 5×5 , and 7×7 . This fusion process significantly enhances the representation of the features.

B. Adaptive Spatio-Temporal Tensor Semantic Awareness

It is essential to explicitly model the spatial-temporal correlations among different frames in echocardiography videos to achieve accurate segmentation results, especially under the scenario of sparse annotation. Current segmentation methods typically use optical flow-based or attention-based modules to exploit pixel-level correlations between frames. However, the presence of speckle noise and limited contrast in echocardiography videos severely degrade image quality, thereby affecting the performance of these approaches. To overcome this challenge, in this section, we propose a novel ASTSA module to utilize the spatio-temporal semantic correlation in video feature tensors. By utilizing this method, we can effectively perceive the global spatio-temporal semantics of video features and characterize the dynamic evolution of segmented objects over time. Before delving into the details, let us first provide the definition of the adaptive t-SVD.

Theorem 1 (Adaptive t-SVD): Given an arbitrary tensor $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, the adaptive t-SVD based on transformation $\mathfrak{S}$ can be factorized as follows,

$$\mathcal{X} = \mathcal{U}_* \mathfrak{S} \mathcal{V}_*^T, \quad (5)$$

where $\mathcal{U} \in \mathbb{R}^{d_1 \times d_1 \times d_3}$ and $\mathcal{V} \in \mathbb{R}^{d_2 \times d_2 \times d_3}$ are orthogonal tensors with respect to the $\mathfrak{S}$ -product. $\mathfrak{S} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$ is a diagonal tensor with each frontal slice being a diagonal matrix.

According to the equations (4) and (5), we can obtain the following expression,

$$\overline{\mathcal{X}_{\mathfrak{S}}} = \overline{\mathcal{U}_{\mathfrak{S}}} \times \overline{\mathcal{S}_{\mathfrak{S}}} \times \overline{\mathcal{V}_{\mathfrak{S}}}^T. \quad (6)$$

By combining equation (6) with equation (1), the n -th frontal slice of $\mathcal{X}_{\mathfrak{S}}$ can be expressed as,

$$\mathcal{X}_{\mathfrak{S}}^{(n)} = \mathcal{U}_{\mathfrak{S}}^{(n)} \mathcal{S}_{\mathfrak{S}}^{(n)} \mathcal{V}_{\mathfrak{S}}^{(n)T}. \quad (7)$$

Therefore, the adaptive t-SVD can be conveniently obtained by computing the matrix SVD of every frontal slice of $\mathcal{X}_{\mathfrak{S}}$ in the transform domain.

In the ASTSA module, we utilize the adaptive t-SVD to decompose the input feature tensor $\mathcal{X}_{in}$ and learn suitable truncation thresholds to preserve the main components of video semantic features. The process can be described as follows,

$$SVT_{\sigma}(\overline{\mathcal{X}_{\mathfrak{S}}}) = \overline{\mathcal{U}_{\mathfrak{S}}} \times \overline{\mathcal{S}_{\mathfrak{S}, \sigma}} \times \overline{\mathcal{V}_{\mathfrak{S}}}^T, \quad (8)$$

where σ represents the singular value threshold vector, and $\overline{\mathcal{S}_{\mathfrak{S}, \sigma}} = \max(\overline{\mathcal{S}_{\mathfrak{S}}} - \sigma, 0)$.

In order to capture spatial-temporal semantic information, we utilize 3D convolutional layers to learn the transformation $\mathfrak{S}$ and the inverse transformation $\tilde{\mathfrak{S}}$, as illustrated in Fig. 2. The input tensor $\mathcal{X}_{in}$ undergoes the transformation $\mathfrak{S}$ to obtain $\mathcal{X}_{in, \mathfrak{S}} \in \mathbb{R}^{C' \times h \times w \times T}$. To improve computational efficiency, we apply t-SVT in parallel to the feature tensor $\mathcal{X}_{in, \mathfrak{S}, c} \in \mathbb{R}^{h \times w \times T}$ based on equation (8), resulting in $\mathcal{X}'_{in, \mathfrak{S}, c}$. Here, $\mathcal{X}_{in, \mathfrak{S}, c}$ represents the c -th channel of $\mathcal{X}_{in, \mathfrak{S}}$, and $c = 1, 2, \dots, C'$. Notably, unlike the conventional SVT [54], our network incorporates a learnable singular value threshold σ_t for each frontal slice. The processed tensors $\mathcal{X}'_{in, \mathfrak{S}, c}$ are then concatenated into a tensor $\mathcal{X}'_{in, \mathfrak{S}} \in \mathbb{R}^{C' \times h \times w \times T}$ along the channel dimension. Finally, we perform the inverse transformation $\tilde{\mathfrak{S}}$ on the tensor $\mathcal{X}'_{in, \mathfrak{S}}$ to obtain the low-rank approximation of tensor $\mathcal{X}_{in}$, denoted as $\mathcal{X}_r \in \mathbb{R}^{C \times h \times w \times T}$.

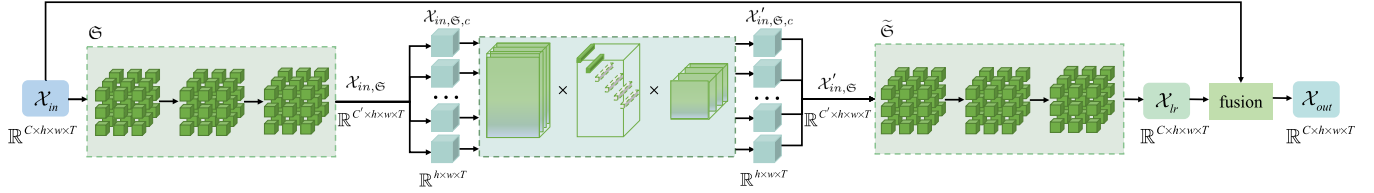


Fig. 2. Our proposed adaptive spatio-temporal tensor semantic awareness (ASTSA) module. The ASTSA extracts the most representative semantic components of dynamic segmentation objects to characterize video feature tensors in the transform domain.

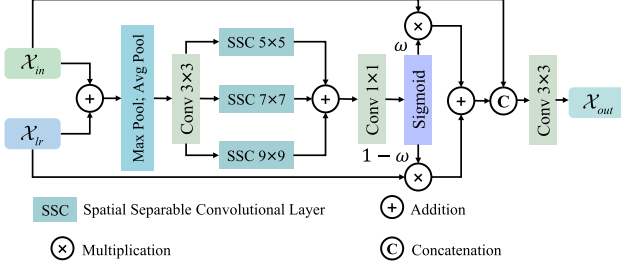


Fig. 3. Illustration of the fusion block, which is responsible for learning the aggregation weights for each input tensor.

The entire process can be expressed as follows,

$$\mathcal{X}_{lr} = \tilde{\mathcal{S}} \circ SVT_{\sigma} \circ \mathcal{S}(\mathcal{X}_{in}), \quad (9)$$

where $\circ$ denotes the composition operator.

To enhance the semantic representation of target tissues, we propose a fusion block that combines the input tensor $\mathcal{X}_{in}$ and the low-rank approximation $\mathcal{X}_{lr}$, as shown in Fig. 3. Specifically, we employ channel pooling operators on the sum of $\mathcal{X}_{in}$ and $\mathcal{X}_{lr}$ to compress the feature tensor and reduce its dimensionality. Additionally, we utilize spatial separable convolutional (SSC) layers [55] with different kernel sizes to extract information with varying receptive fields. These extracted features are then combined and processed using the Sigmoid operator to generate dynamic aggregation weights for the input tensors, representing their importance in the final representation. Finally, we concatenate $\mathcal{X}_{lr}$ with the weighted sum of $\mathcal{X}_{in}$ and $\mathcal{X}_{lr}$. We then obtain the final output of the fusion block by applying a convolutional layer, denoted as $\mathcal{X}_{out}$.

C. Memory Flow

In the mentioned ASTSA module, we aim to effectively capture the evolution of dynamic targets in videos by exploiting the spatio-temporal semantic correlation of video feature tensors. However, this approach inevitably loses some detailed but crucial features required for accurate segmentation, such as edge information. To overcome this limitation, inspired by [56], we propose the MF module, which calculates non-local pixel-level similarity between two adjacent frames to propagate both semantics and mask information, thereby enhancing the accuracy of segmentation.

Fig. 4 illustrates the overview of our proposed MF module. As shown in Fig. 4 (a), we utilize the feature of the previous frame F_{t-1} , the corresponding segmentation result M_{t-1} , and the feature of the current frame (F_t and F'_t) to

generate two important components: **Key** (K) and **Value** (V). The K component encodes semantic information to calculate the affinity matrix to assign weights to V at different pixel locations, while the V component stores detailed information for generating segmentation results. Specifically, we feed F_{t-1} and F_t into ProjectorK to generate K_{t-1} and K_t , respectively. In addition, we use the MaskConv block to generate V_{t-1} by taking F_{t-1} and M_{t-1} as inputs, and we use ProjectorV to generate V_t from F'_t . Here, $K_{t-1}, K_t \in \mathbb{R}^{C^k \times h \times w}$ and $V_{t-1}, V_t \in \mathbb{R}^{C^v \times h \times w}$, where C^k and C^v represent the number of channels of K and V , respectively. Both ProjectorK and ProjectorV consist of a 3×3 convolutional layer.

To effectively incorporate both semantics and mask information, we employ the MaskConv block to generate V_{t-1} , as illustrated in Fig. 4 (b). In the MaskConv block, we draw inspiration from the multi-head attention mechanism [57] and divide the feature F_{t-1} into multiple groups. The feature in each group is concatenated with M_{t-1} , which is resized to the same size as the feature. Subsequently, we apply basic unit (BU) blocks with varying convolutional kernel sizes (1, 3, 5, and 7) to the different groups to extract information at different scales. Each BU block consists of two convolutional layers and a ReLU nonlinear activation function. Finally, we concatenate the groups along the channel dimension and utilize a convolutional layer with kernel size of 1 to achieve interaction between features at different scales, obtaining V_{t-1} .

In order to effectively handle variations between frames, we calculate affinity metrics at multiple scales. Specifically, we utilize MaxPool operators to downsample K_{t-1} , K_t , and V_{t-1} , obtaining K_{t-1}^l, K_t^l , and V_{t-1}^l respectively, where $l \in \{1, 2, 3\}$ represents the scale index. The affinity matrix is computed as depicted in Fig. 4 (c). We first reshape K_{t-1}^l and K_t^l into matrices of dimension $\mathbb{R}^{C^k \times h^l w^l}$, and we then calculate the normalized affinity matrix $A^l \in \mathbb{R}^{h^l w^l \times h^l w^l}$ as follows,

$$A_{h,j}^l = \frac{\exp\left(c\left(K_{t-1,h}^l, K_{t,j}^l\right)\right)}{\sum_h \left(\exp\left(c\left(K_{t-1,h}^l, K_{t,j}^l\right)\right)\right)}, \quad (10)$$

where h and j represent the location indexes. The function $c(\cdot, \cdot)$ denotes the L2 similarity measure, which can be computed efficiently as demonstrated in [58]. Based on A^l , we obtain $V_R^l \in \mathbb{R}^{C^v \times h^l \times w^l}$ as follows,

$$V_R^l = V_{t-1}^l A^l. \quad (11)$$

Next, we fuse V_R^l across all scales, as shown in Fig. 5. First, we upsample V_R^2 and V_R^3 to the size of V_R^1 and apply

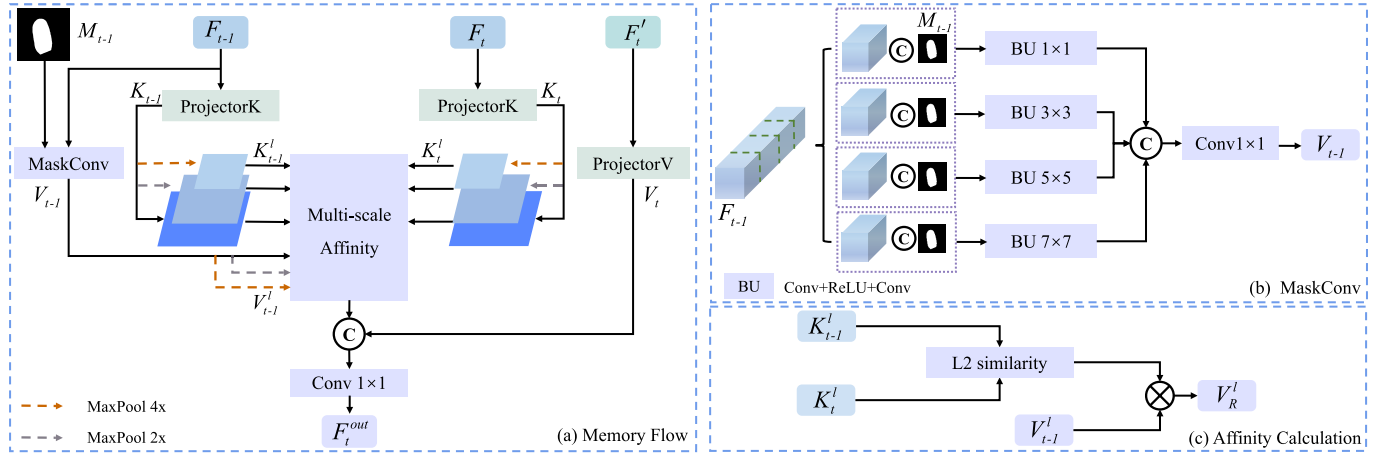


Fig. 4. Overview of the memory flow (MF). (a) Structure of the MF. (b) Structure of the MaskConv block in the MF. (c) Illustration of affinity calculation at each scale l . The MF module propagates relevant information from the previously segmented frame to the current frame to be segmented.

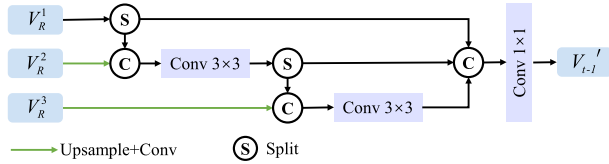


Fig. 5. The fusion process of the multi-scale affinity block in the memory flow module.

convolution to reduce the number of channels. We then split V_R^1 along the channel dimension, concatenate one part with V_R^2 , and pass it through a convolutional layer. The result is split again, and part of it is concatenated with V_R^3 , followed by another convolutional layer. Subsequently, we concatenate the processed features to produce the fused result, V_{t-1}^f , which is then concatenated with V_t and fed into a 1×1 convolutional layer to generate the final output feature $F_t^{out} \in \mathbb{R}^{C \times h \times w}$. The F_t^{out} integrates information from both the previous frame and the current frame, and it is fed into the decoder to produce the segmentation result M_t for the current frame.

D. Training Loss Function

In medical image segmentation tasks, cross-entropy (CE) loss [59] and Dice loss [60] are commonly used as loss functions. The CE loss measures the disparity between the distribution of labels and the network's predictions, which can be described as follows,

$$\mathcal{L}_{CE} = - \sum G(i, j) \cdot \log P(i, j) + (1 - G(i, j)) \cdot \log(1 - P(i, j)), \quad (12)$$

where $P(i, j)$ is the prediction result at pixel (i, j) , and $G(i, j)$ is the ground truth label at pixel (i, j) .

Dice loss compares the similarity between the actual segmented area and the predicted segmented area, which is defined as follows,

$$\mathcal{L}_{Dice} = 1 - \frac{2|G \cap P|}{|G| + |P|}. \quad (13)$$

In our work, we adopt a hybrid loss that combines CE loss and Dice loss to supervise the SAMF-Net. The loss function can be written as follows,

$$\mathcal{L} = 0.5 * \mathcal{L}_{Dice}(P, G) + 0.5 * \mathcal{L}_{CE}(P, G). \quad (14)$$

E. Implementation Details

We developed our proposed SAMF-Net utilizing the Pytorch deep-learning framework and implemented it on an NVIDIA Tesla A100 GPU. In our network, we used the YoloV8 feature extraction network [61] as the backbone in the encoder, which was pre-trained on the COCO dataset [62]. Our decoder structure was consistent with that of the [56]. In the ASTSA module, we set the value of C' as 32. In the MF module, we set the values of C^k and C^v as 64 and 128, respectively. The learning rate was initially set to 0.0001, and we exponentially decayed it for every epoch. To train the proposed network, we used the AdamW optimizer [63] with a weight decay parameter of 0.0005 for a total of 70 epochs.

IV. MATERIALS AND EXPERIMENTS

A. Datasets and Metrics

In this study, we assessed the effectiveness of our proposed semi-supervised echocardiography video segmentation method using two public datasets and a private dataset.

1) **Public Datasets:** The first dataset, CAMUS [2], comprises echocardiography video data from 500 patients at the University Hospital of St. Etienne in France. The dataset includes apical two-chamber and apical four-chamber views for each patient, collected using GE Vivid E95 ultrasound scanners, with a total of 1000 echocardiography sequences. Each sequence consists of an average of 20 frames with varying sizes and only two frames (end-systole and end-diastole) are manually labeled by an experienced cardiologist. Statistically, half of the patients exhibit a left ventricular ejection fraction lower than 45%, indicating a pathological risk. On the other hand, 19% of the sequences have poor image quality, which poses challenges for segmentation.

The second dataset, EchoNet-Dynamic [64], consists of 10030 video sequences collected at Stanford University Hospital. The dataset was acquired using five different ultrasound scanners and includes data from both cardiac patients and healthy volunteers. Each echocardiography video covers multiple cardiac cycles and contains approximately 200 frames with a resolution of 112×112 pixels. Similar to the CAMUS dataset, only two frames (end-systole and end-diastole) are manually labeled. The dataset has been pre-divided into training (7465 videos), validation (1288 videos), and testing (1277 videos).

2) Private Dataset: To further validate the clinical applicability of the proposed method, we collected a private dataset from multiple hospitals in China, including Lanzhou University First Hospital, Lanzhou, Gansu Province, China, and Jiaxing Second Hospital, Jiaxing, Zhejiang Province, China, and so on. The dataset was collected using various ultrasound scanners, including CX 50, EPIQ 5, and EPIQ 7 from Philips, as well as LOGIQE 11, Vivid E9, and Vivid E95 from GE. To protect the privacy of patients and comply with ethical guidelines, all personal information was removed and the data was anonymized. The echocardiography videos were acquired in the standard apical two-chamber plane for 550 patients. Each video contains 2-4 cardiac cycles, with 40 to 200 frames and varying spatial resolutions. For each video, two sonographers with 3-5 years of experience marked the LV contours on the ED and ES frames within only one cardiac cycle, which were then double-checked by a senior sonographer with over 10 years of experience.

We conducted 5-fold cross-validation on the CAMUS and Private datasets. Specifically, for the CAMUS dataset, we equally divided the dataset into 5 folds to ensure that cases with poor image quality and low ejection fraction were evenly distributed across all folds. For the Private dataset, we considered variations in image quality and distributed lower-quality images evenly across the folds. During cross-validation, we trained the model using data from 4 folds and tested it on the remaining fold, repeating this process for each fold. The image size of the CAMUS and Private datasets was standardized to 256×256 using cubic interpolation. For the larger EchoNet-Dynamic dataset, we followed the predefined data division, kept the image size at 112×112 , and conducted experiments five times with different random seeds.

To avoid overfitting and improve the generalization of our proposed method, we performed data augmentations to enrich the diversity of training samples, including random horizontal flipping, random scale, random rotation, random contrast, and gamma enhancement with a probability of 0.5 for each operation. In each sequence, we sampled a total of 10 frames that were evenly distributed, starting from ED and ending with ES. It is important to note that only the ED and ES frames had corresponding labels.

We employed three evaluation metrics to assess the accuracy of the segmentation, including the Dice Similarity Coefficient (DC) [66], Sensitivity (Sen) [67], and Average Surface Distance (ASD) [68]. The DC and Sen metrics were used to evaluate the precision of the segmentation results, while the

ASD metric was used to measure the distance between the predicted boundary and the true boundary.

B. Comparison With the State-of-the-Art Methods

We compared our proposed SAMF-Net with ten state-of-the-art methods, including STM [56], DAEF [12], Swin-Unet [65], LV-Net [13], SSCF [22], AAU-Net [11], CLAS [26], PKEcho-Net [19], MemSAM [38], and SimLVSeg [20]. To ensure a fair comparison, we maintained consistent settings for data augmentation and learning rate adjustment across all methods. Additionally, all experiments were conducted in the same computing environment.

Fig. 6 shows the segmentation results obtained from different methods. The first three rows of the figure display the results of one frame from the CAMUS dataset, while the subsequent three rows present the results of one frame from the EchoNet-Dynamic dataset. The figure displays a series of echocardiography frames and the corresponding segmentation results produced by various methods, including STM, DAEF, Swin-Unet, LV-Net, SSCF, AAU-Net, CLAS, PKEcho-Net, MemSAM, SimLVSeg, and our proposed SAMF-Net. In order to visually illustrate the discrepancy between the segmentation results and the provided labels, we used green, red, and yellow colors to represent the labels, segmentation results, and their overlapping areas, respectively.

The echocardiography frames in Fig. 6 exhibit several challenging cases with speckle noise, blurred edges, and low contrast of the left ventricle (LV). Single-frame segmentation methods do not utilize temporal information and usually exhibit poor segmentation results, especially when boundaries are blurred. SSCF employs an adaptive spatio-temporal semantic calibration module that warps adjacent frame features to the target frame using inter-frame attention, but it cannot capture correlations across the entire video. CLAS incorporates optical flow estimation to capture alterations in the heart's shape and appearance. However, the presence of speckle noise complicates pixel-level optical flow calculation. Both STM and MemSAM process echocardiography videos in a sequential frame-by-frame manner, limiting their spatio-temporal modeling capabilities and affecting segmentation accuracy. In contrast, our proposed SAMF-Net performs adaptive tensor singular value decomposition on the video feature tensor based on the low-rank prior of echocardiography video features, effectively capturing the key semantics of the moving target and achieving satisfactory segmentation results.

In addition to visual comparisons, we conducted quantitative statistical comparisons of different methods using the CAMUS dataset and the EchoNet-Dynamic dataset. The mean and standard deviation of different runs are presented in Table I. The results show that our proposed method consistently achieves the highest accuracy and lowest standard deviation across the DC, Sen, and ASD metrics on both datasets, demonstrating its robustness and reliability compared to the other state-of-the-art methods. By integrating the ASTSA and MF modules, we can not only fully exploit the spatio-temporal semantic information of dynamic tissues, but also mitigate the impact of speckle noise to improve the accuracy of segmentation results.

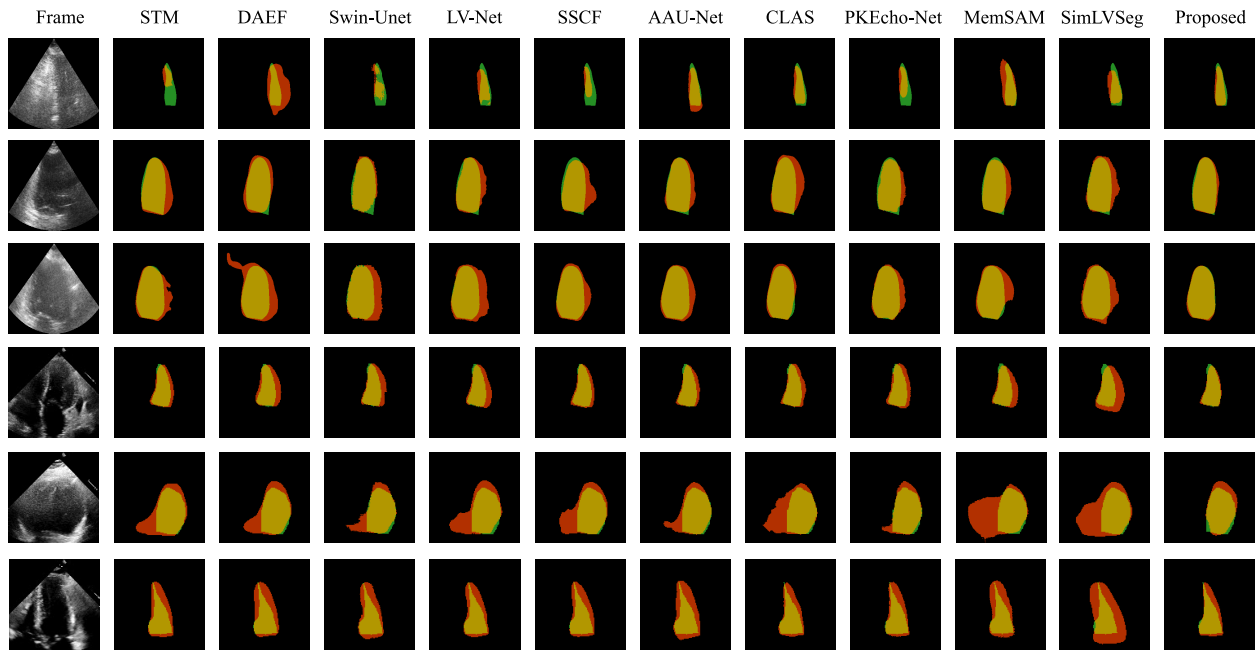


Fig. 6. Visual comparison of segmentation results obtained from different methods on the CAMUS and EchoNet-Dynamic public datasets. The first three rows illustrate the results of one frame from the CAMUS dataset, and the subsequent three rows present the results of one frame from the EchoNet-Dynamic dataset. The columns from left to right consist of the frames and the corresponding segmentation results obtained by the following methods: STM, DAEF, Swin-Unet, LV-Net, SSCF, AAU-Net, CLAS, PKEcho-Net, MemSAM, SimLVSeg, and our proposed SAMF-Net. The green, red, and yellow regions represent the ground truth, prediction, and their overlapping regions, respectively.

TABLE I
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE CAMUS AND EchoNet-DYNAMIC PUBLIC DATASETS (MEAN $\pm$ STANDARD DEVIATION OF 5 RUNS)

Method	Venue & Year	CAMUS			EchoNet-Dynamic		
		DC (%)	Sen (%)	ASD	DC (%)	Sen (%)	ASD
STM [56]	ICCV 2019	92.14 $\pm$ 5.40	92.41 $\pm$ 6.37	2.78 $\pm$ 1.40	91.27 $\pm$ 5.37	91.10 $\pm$ 6.17	1.29 $\pm$ 0.52
DAEF [12]	MIA 2021	92.51 $\pm$ 4.97	92.60 $\pm$ 6.42	2.58 $\pm$ 1.31	91.63 $\pm$ 4.06	90.99 $\pm$ 6.59	1.21 $\pm$ 0.50
Swin-Unet [65]	ECCV 2022	90.72 $\pm$ 5.45	92.32 $\pm$ 6.63	2.83 $\pm$ 1.38	91.57 $\pm$ 4.12	91.58 $\pm$ 6.03	2.35 $\pm$ 1.25
LV-Net [13]	TUFFC 2022	91.47 $\pm$ 5.59	92.01 $\pm$ 7.42	2.98 $\pm$ 1.73	91.53 $\pm$ 4.06	91.76 $\pm$ 5.81	1.21 $\pm$ 0.64
SSCF [22]	MIA 2022	93.21 $\pm$ 5.02	93.29 $\pm$ 6.55	2.35 $\pm$ 1.31	92.15 $\pm$ 3.88	93.21 $\pm$ 6.13	1.18 $\pm$ 0.54
AAU-Net [11]	TMI 2022	93.28 $\pm$ 4.72	93.42 $\pm$ 5.80	2.64 $\pm$ 1.77	92.21 $\pm$ 3.79	93.15 $\pm$ 5.86	1.12 $\pm$ 0.52
CLAS [26]	MIA 2023	93.19 $\pm$ 4.36	93.38 $\pm$ 5.47	2.53 $\pm$ 1.47	92.13 $\pm$ 3.79	93.31 $\pm$ 5.39	1.25 $\pm$ 0.61
PKEcho-Net [19]	AAAI 2023	93.32 $\pm$ 4.40	93.47 $\pm$ 5.70	2.45 $\pm$ 1.34	92.27 $\pm$ 3.72	93.01 $\pm$ 6.04	1.29 $\pm$ 0.58
MemSAM [38]	CVPR 2024	92.89 $\pm$ 5.82	93.19 $\pm$ 6.60	2.67 $\pm$ 1.87	91.93 $\pm$ 5.44	93.48 $\pm$ 5.68	1.43 $\pm$ 1.66
SimLVSeg [20]	UMB 2024	92.78 $\pm$ 4.59	93.16 $\pm$ 5.67	2.80 $\pm$ 1.49	91.84 $\pm$ 5.73	93.30 $\pm$ 5.82	2.56 $\pm$ 1.72
Proposed	-	94.23$\pm$3.41	94.51$\pm$4.71	1.97$\pm$0.91	92.92$\pm$3.31	94.20$\pm$5.01	0.99$\pm$0.43

To further validate the effectiveness of our proposed method, we conducted experiments using the Private dataset, which consists of authentic clinical data. The segmentation results obtained from different methods are displayed in Fig. 7. Despite the absence of distinct delineations between the LV and the background tissue in frames, our proposed method exhibits the closest segmentation results to the ground truth labels. The quantitative comparison results of the Private dataset are displayed in Table. II. It is demonstrated that the SAMF-Net achieves the highest accuracy compared to the other state-of-the-art echocardiography video segmentation methods.

We conducted the Wilcoxon rank sum test [69] for each metric to evaluate the significance of the performance improvements. The results for all datasets are summarized in Table III, where P-value indicates the significance level of the statistical test. A P-value below the standard significance threshold

(typically 0.05) represents a statistically significant difference between the two experimental results [2]. As shown in Table III, all P-values are below 0.05, confirming that our method's improvements on all datasets are statistically significant.

C. Out-of-Distribution Testing

The generalization capability of algorithms in medical applications is essential, considering the variability in equipment and data collection processes across hospitals. Therefore, we conducted out-of-distribution (OOD) testing [70] to assess the generalization performance of our method. Specifically, we evaluated our method and the comparison methods on an additional private OOD dataset, using models trained on the CAMUS dataset with 5-fold cross-validation.

The OOD dataset consists of apical four-chamber echocardiography sequences from 150 patients, collected using Philips

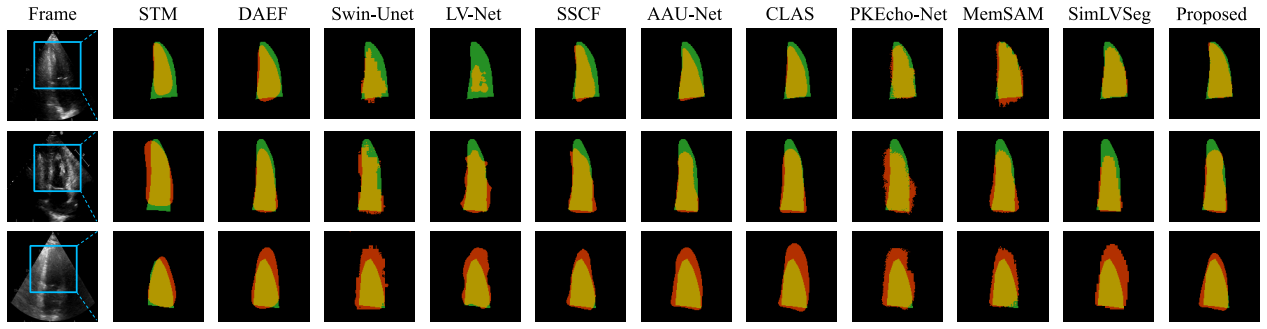


Fig. 7. Visual comparison of segmentation results obtained from different methods on the Private dataset. The columns from left to right consist of the frames and the corresponding segmentation results obtained by the following methods: STM, DAEF, Swin-Unet, LV-Net, SSCF, AAU-Net, CLAS, PKEcho-Net, MemSAM, SimLVSeg, and our proposed SAMF-Net. We cropped the visualization results based on the label regions to facilitate observing the differences between different methods. The green, red, and yellow regions represent the ground truth, prediction, and their overlapping regions, respectively.

TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE PRIVATE DATASET (MEAN $\pm$ STANDARD DEVIATION OF 5 RUNS)

Method	Venue & Year	DC (%)	Private Sen (%)	ASD
STM [56]	ICCV 2019	90.53 $\pm$ 4.28	89.15 $\pm$ 7.26	2.49 $\pm$ 1.38
DAEF [12]	MIA 2021	91.16 $\pm$ 4.10	89.90 $\pm$ 7.68	2.45 $\pm$ 0.90
Swin-Unet [65]	ECCV 2022	90.35 $\pm$ 3.67	89.97 $\pm$ 6.88	2.74 $\pm$ 0.85
LV-Net [13]	TUFFC 2022	90.86 $\pm$ 3.51	90.07 $\pm$ 6.60	2.63 $\pm$ 0.98
SSCF [22]	MIA 2022	91.34 $\pm$ 3.20	91.33 $\pm$ 5.69	2.37 $\pm$ 0.69
AAU-Net [11]	TMI 2022	91.61 $\pm$ 3.70	91.36 $\pm$ 5.84	2.47 $\pm$ 1.11
CLAS [26]	MIA 2023	91.40 $\pm$ 3.42	91.18 $\pm$ 5.40	2.33 $\pm$ 0.79
PKEcho-Net [19]	AAAI 2023	91.65 $\pm$ 3.18	90.98 $\pm$ 5.66	2.52 $\pm$ 0.76
MemSAM [38]	CVPR 2024	90.90 $\pm$ 3.78	91.32 $\pm$ 5.95	2.59 $\pm$ 1.39
SimLVSeg [20]	UMB 2024	90.94 $\pm$ 3.51	90.21 $\pm$ 7.04	2.61 $\pm$ 0.74
Proposed	-	92.38$\pm$2.82	92.19$\pm$5.12	2.13$\pm$0.51

TABLE IV

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE OUT-OF-DISTRIBUTION (OOD) DATASET

Method	Venue & Year	DC (%)	OOD Sen (%)	ASD
STM [56]	ICCV 2019	88.56 $\pm$ 5.32	91.13 $\pm$ 6.32	3.76 $\pm$ 2.38
DAEF [12]	MIA 2021	87.13 $\pm$ 7.61	89.90 $\pm$ 7.71	4.53 $\pm$ 2.55
Swin-Unet [65]	ECCV 2022	88.06 $\pm$ 5.75	91.74 $\pm$ 5.69	4.07 $\pm$ 2.71
LV-Net [13]	TUFFC 2022	88.46 $\pm$ 5.65	91.20 $\pm$ 6.85	4.24 $\pm$ 2.55
SSCF [22]	MIA 2022	89.75 $\pm$ 5.45	91.15 $\pm$ 6.91	4.77 $\pm$ 3.72
AAU-Net [11]	TMI 2022	90.83 $\pm$ 7.58	92.26 $\pm$ 8.15	3.52 $\pm$ 4.60
CLAS [26]	MIA 2023	89.66 $\pm$ 5.62	90.89 $\pm$ 6.33	4.65 $\pm$ 3.43
PKEcho-Net [19]	AAAI 2023	90.06 $\pm$ 5.43	93.06 $\pm$ 5.37	3.38 $\pm$ 2.57
MemSAM [38]	CVPR 2024	90.34 $\pm$ 5.91	90.86 $\pm$ 6.36	3.61 $\pm$ 2.89
SimLVSeg [20]	UMB 2024	89.53 $\pm$ 6.60	90.97 $\pm$ 6.95	4.00 $\pm$ 3.40
Proposed	-	92.08$\pm$4.07	93.52$\pm$5.18	2.93$\pm$1.84

TABLE III

P-VALUES OF WILCOXON RANK-SUM TEST BETWEEN OUR METHOD AND THE OTHER STATE-OF-THE-ART METHODS ON DIFFERENT METRICS

Method	CAMUS			EchoNet-Dynamic			Private		
	DC	Sen	ASD	DC	Sen	ASD	DC	Sen	ASD
vs. STM	5e-5	0.046	6e-4	1e-5	5e-5	7e-4	0.009	0.015	0.018
vs. DAEF	1e-5	0.004	2e-6	3e-4	8e-6	9e-4	0.029	2e-5	0.046
vs. Swin-Unet	6e-4	1e-4	1e-5	1e-6	3e-5	2e-5	5e-4	0.013	5e-4
vs. LV-Net	3e-6	2e-5	4e-7	6e-6	5e-4	3e-4	0.001	0.011	8e-4
vs. SSCF	7e-5	0.001	7e-7	1e-4	3e-6	5e-5	0.016	0.020	0.047
vs. AAU-Net	9e-5	0.005	1e-6	0.003	0.026	0.003	0.025	7e-4	0.013
vs. CLAS	0.009	0.045	0.001	0.017	1e-5	0.002	0.007	0.012	4e-5
vs. PKEcho-Net	0.004	1e-5	4e-6	2e-7	8e-5	3e-5	0.007	0.019	0.002
vs. SimLVSeg	6e-4	4e-5	6e-4	3e-5	1e-6	1e-4	0.003	2e-4	0.008
vs. MemSAM	3e-7	0.020	3e-5	5e-7	6e-6	1e-5	8e-4	4e-6	9e-5

EPIQ 7C and GE Vivid E95. Each sequence contains at least 20 frames with a resolution of 800×600 pixels. The frames are downsampled to 256×256 pixels using cubic interpolation. The dataset was sourced from multiple hospitals, providing a diverse clinical population ideal for generalizability testing. The mean and standard deviation of the 5 evaluations are presented in Table IV. The results demonstrate that our proposed method achieves a Dice coefficient improvement of 1.25% to 4.95% on the OOD dataset compared to the other methods, while also reducing the standard deviation by 1.25% to 3.54%. This test further validates our method's robustness and generalization on the unseen dataset, highlighting its applicability in clinical settings.

D. Speed Analysis

In echocardiography video segmentation tasks, it is crucial to ensure the real-time performance of the network. In this

TABLE V

METHODS EFFICIENCY COMPARISON AT A RESOLUTION OF 256×256 , REPORTING THE NUMBER OF MODEL PARAMETERS (PARAMS), FRAMES PER SECOND (FPS), AND AVERAGE FLOATING POINT OPERATIONS PER SECOND (FLOPS) PER FRAME

Method	Params(M)	FPS	FLOPs(G)
STM	38.9	158	26
DAEF	69.5	153	103
Swin-Unet	60.1	85	98
LV-Net	1.6	125	32
SSCF	57.7	77	63
AAU-Net	41.9	95	71
CLAS	19.1	90	65
PKEcho-Net	25.7	230	11
SimLVSeg	18.8	110	10
MemSAM	39.7	34	146
Proposed	21.4	160	40

section, we conducted a comparative analysis of the network performance to evaluate the effectiveness of our method compared to the other state-of-the-art methods. Table V presents the number of model parameters (Params), frames per second (FPS), and floating point operations per second (FLOPs) for all methods. Our approach has fewer parameters compared to the other methods, requiring only 21.4 M learnable parameters. The inference speed of our approach is 160 FPS, achieving real-time segmentation in our experimental environment. In terms of computational complexity, our method has an average of 40 G FLOPs per frame, which is more efficient than most of the comparative methods.

TABLE VI

QUANTITATIVE COMPARISON OF ABLATION EXPERIMENTS ON THE CAMUS DATASET. THE BASELINE MODEL ONLY HAS AN ENCODER AND DECODER. MSF: MULTI-SCALE FUSION MODULE. ASTSA: ADAPTIVE SPATIO-TEMPORAL TENSOR SEMANTIC AWARENESS MODULE. MF: MEMORY FLOW MODULE

Method	DC(%)	Sen(%)	ASD
Baseline	93.17±5.53	92.91±6.67	2.50±1.60
Baseline+MSF	93.77±4.43	93.93±5.61	2.29±1.12
Baseline+ASTSA	93.90±3.80	94.14±5.14	2.15±1.08
Baseline+MF	93.71±4.02	93.86±5.54	2.38±1.25
Baseline+MSF+ASTSA	94.07±3.51	94.37±4.90	1.98±0.99
Baseline+MSF+MF	93.94±3.72	94.11±5.06	2.12±1.02
Baseline+ASTSA+MF	93.99±3.63	94.32±4.88	2.05±1.29
Proposed	94.23±3.41	94.51±4.71	1.97±0.91

E. Ablation Experiments

We conducted ablation experiments with 5-fold cross-validation on the CAMUS dataset to demonstrate the contributions of the proposed multi-scale fusion (MSF) module, adaptive spatio-temporal tensor semantic awareness (ASTSA) module, and memory flow (MF) module. We implemented a baseline method that consists of a shared encoder and decoder. Subsequently, we incorporated the MSF, ASTSA, and MF modules into the Baseline approach, respectively, resulting in six variations: Baseline+MSF, Baseline+ASTSA, Baseline+MF, Baseline+MSF+ASTSA, Baseline+MSF+MF, and Baseline+ASTSA+MF. Finally, we combined all three modules to obtain the complete proposed method. We report the mean and standard deviation of 5 folds in Table VI. Compared to the Baseline, the inclusion of the MSF module increased the Dice coefficient by 0.60% and reduced the standard deviation by 1.1%, highlighting the benefits of incorporating multi-scale features. The integration of the ASTSA module enabled effective utilization of video information and the acquisition of compact semantic representations, resulting in a significant improvement of 0.73% in the Dice coefficient. Additionally, the MF module facilitated the propagation of valuable information from previously segmented frames to the subsequent frame, achieving improved segmentation performance compared to the Baseline. These results demonstrate that each individual component contributes to improved performance over the Baseline, while the combination of MSF, ASTSA, and MF modules further enhances the model's overall performance.

We also conducted the Wilcoxon rank sum test on all metrics as presented in Table VII. All P-values are below 0.05, confirming that the performance gains of the proposed components are statistically significant.

F. Impact of Hyperparameters

To evaluate the impact of the hyperparameters C' , C^k , and C^v on model performance, we conducted experiments on the CAMUS dataset. Specifically, we set C' to 16, 24, 32, 40, and 48, C^k to 32, 48, 64, 80, and 96, and C^v to 64, 96, 128, 160, and 192, respectively. The experimental results are presented in Fig. 8. We observe that the Dice coefficients improve as C' increases up to 32 and C^k up to 64. As C^v increases to 128, the

TABLE VII

P-VALUES OF WILCOXON RANK-SUM TEST BETWEEN OUR METHOD AND THE ABLATION STUDY OF THE PROPOSED MODULE ON THE CAMUS DATASET

Method	DC	Sen	ASD
vs. Baseline	3e-5	3e-6	1e-5
vs. Baseline+MSF	8e-4	5e-4	3e-5
vs. Baseline+ASTSA	0.010	0.004	2e-5
vs. Baseline+MF	1e-4	2e-5	3e-4
vs. Baseline+MSF+ASTSA	0.028	0.012	4e-6
vs. Baseline+MSF+MF	0.002	7e-4	1e-4
vs. Baseline+ASTSA+MF	0.003	9e-4	5e-5

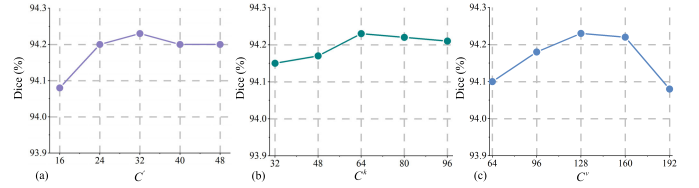


Fig. 8. Impact of hyperparameters C' , C^k , and C^v on Dice coefficient performance on the CAMUS dataset. (a) Impact of C' . (b) Impact of C^k . (c) Impact of C^v .

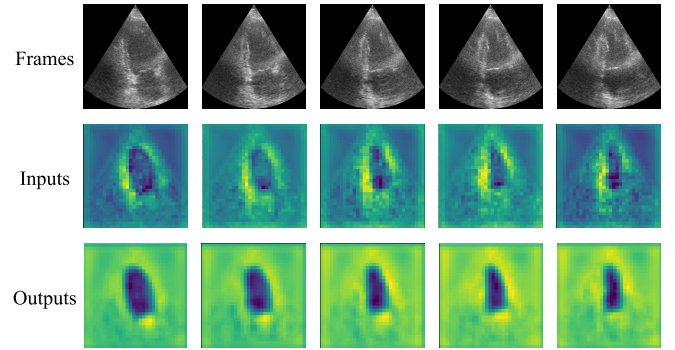


Fig. 9. Visualization comparison of feature maps before and after the ASTSA module processing. The first row shows equally spaced frames from an echocardiography video. The second row shows a channel of the input feature maps for each frame in the ASTSA module. The third row shows the corresponding channel of the output feature maps for each frame after processed by the ASTSA module.

Dice coefficient also improves. However, further increases in C^v lead to a performance degradation of about 0.15%. Based on these findings, we select C' as 32, C^k as 64, and C^v as 128, as these values provide a balance between segmentation accuracy and model complexity.

G. Visualization of The ASTSA Module's Effect

The proposed ASTSA module decomposes and processes the echocardiography video feature tensor to reduce feature redundancy and obtain a low-rank approximation of the feature tensor. In order to demonstrate the effect of the ASTSA module, which was illustrated in Fig. 2, we compare the input and output of the ASTSA module based on the CAMUS dataset. As shown in Fig. 9, the first row displays 5 equally spaced frames from the video, all belonging to the same cardiac cycle of the same patient. The second row shows a channel of the input feature maps for each frame in the ASTSA module, while the third row depicts the corresponding

feature maps after being processed by the ASTSA module. It is evident from the figure that the input feature maps contain a lot of noise, and the boundaries between the LV and the background are blurred. However, these issues are effectively alleviated through the application of the ASTSA module. Specifically, the ASTSA module plays a crucial role in integrating spatio-temporal semantic features from all frames within the echocardiography video. Based on the low-rank prior of the video feature tensor, the ASTSA module utilizes adaptive t-SVD to capture the most representative semantic feature components. This process enables efficient characterization of video features by removing redundant features and eliminating the influence of speckle noises. As a result, it improves the distinguishability of features between LV and background, improving the accuracy of echocardiography video segmentation.

V. CONCLUSION

In this paper, we proposed a novel approach for semi-supervised segmentation in echocardiography videos based on the low-rank prior of video feature tensors. We exploited the spatio-temporal semantic correlation of video feature tensors to capture the temporal evolution of the dynamic target tissue, effectively overcoming the limitations of sparse annotation and improving the accuracy of video segmentation. In the proposed adaptive spatio-temporal tensor semantic awareness module, we performed adaptive tensor singular value decomposition on the video feature tensor and learned appropriate thresholds to retain key semantic feature components, resulting in a compact and continuous video feature representation. In addition, we introduced a memory flow module to propagate useful information between adjacent frames, which can effectively respond to inter-frame changes and further enhance the accuracy of video segmentation. We conducted extensive experimental assessments on both public datasets and a privately collected dataset. The results demonstrated the superior performance of our method compared to state-of-the-art methods for echocardiography video segmentation.

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